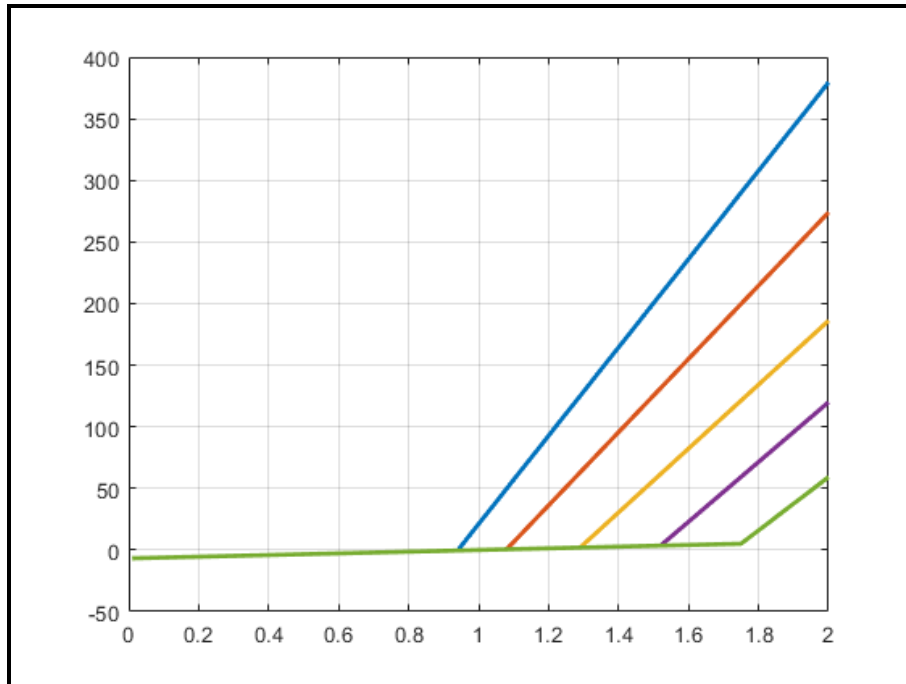


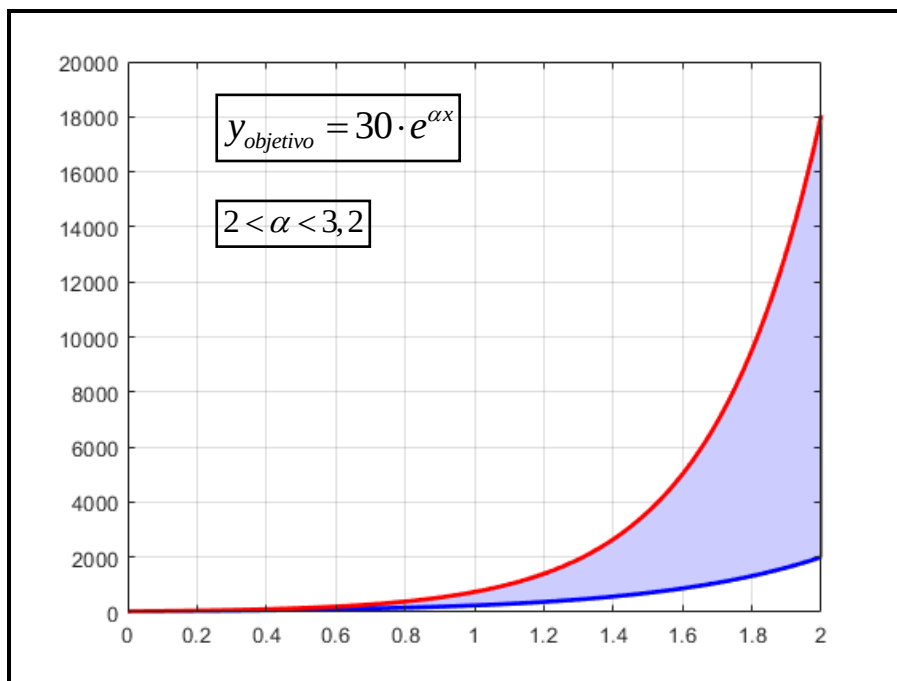
Aproximación de la función objetivo con una combinación lineal de coeficientes enteros para 5 funciones base, realizado con una Red Neuronal entrenada.

Los datos de entrenamiento se logran buscando los valores óptimos con el método de los Multiplicadores de Lagrange (reales). Luego se entrenó a una Red Neuronal con datos generados por el método numérico (reales). Luego se compararon las predicciones del método numérico y los de la red neuronal (reales). Luego se redondearon las predicciones de la red neuronal (enteros), y se generaron las curvas de aproximación finales.

Funciones Base nuevas



Rango de variación de las funciones objetivo para aproximaciones con $R^2 > 0,95$



Los algoritmos de aproximación numérica y de red neuronal se realizaron con números reales **R**

TABLA 1: COEFICIENTES DE LAGRANGE (REALES)

α	c1	c2	c3	c4	c5	R ²
2.00	3.0327	0.0000	0.0000	2.5631	0.8720	0.94842235
2.20	3.9608	0.0000	0.0000	5.0407	2.5870	0.96383581
2.50	5.8935	0.0000	0.0000	11.8558	7.8189	0.97630930
2.70	7.6612	0.0000	0.0000	19.7488	14.5284	0.98052212
2.75	8.1435	0.0000	0.7906	20.0000	19.5073	0.98113746
2.80	8.2183	0.0000	3.7770	20.0000	20.0000	0.98135998
3.00	8.3016	0.0000	20.0000	20.0000	20.0000	0.97741570
3.20	4.0686	20.0000	20.0000	20.0000	20.0000	0.96182896

TABLA 2: COEFICIENTES DE LA RN (REALES)

α	c1	c2	c3	c4	c5	R ²
2.00	3.0470	-0.0872	-0.0658	2.6172	0.7703	0.94820094
2.20	3.9544	-0.0235	-0.0415	5.1199	2.5598	0.96387378
2.50	5.8920	0.0477	-0.0329	11.9184	7.7930	0.97627335
2.70	7.6739	0.0574	-0.0224	19.7937	14.5825	0.98047911
2.75	8.1321	0.0591	0.8758	19.9855	19.5213	0.98109536
2.80	8.2062	0.0363	3.7818	20.0059	20.1106	0.98136080
3.00	8.3420	0.0446	19.8157	19.9911	20.0411	0.97740998
3.20	3.2494	20.5746	19.9206	20.0577	19.7431	0.96146192

TABLA 3: COEFICIENTES DE LA RN (REDONDEADOS A ENTEROS)

α	c1	c2	c3	c4	c5	R ²
2.00	3	0	0	3	1	0.94752125
2.20	4	0	0	5	3	0.96358894
2.50	6	0	0	12	8	0.97585243
2.70	8	0	0	20	15	0.97880898
2.75	8	0	1	20	20	0.98111273
2.80	8	0	4	20	20	0.98124064
3.00	8	0	20	20	20	0.97713661
3.20	3	21	20	20	20	0.96173119

=====

RESUMEN DE R² - COMPARATIVA

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α	R ² Lagrange	R ² RN real	R ² RN redondeada
2.00	0.94842235	0.94820094	0.94752125
2.20	0.96383581	0.96387378	0.96358894
2.50	0.97630930	0.97627335	0.97585243
2.70	0.98052212	0.98047911	0.97880898
2.75	0.98113746	0.98109536	0.98111273
2.80	0.98135998	0.98136080	0.98124064
3.00	0.97741570	0.97740998	0.97713661
3.20	0.96182896	0.96146192	0.96173119

=====

ESTADÍSTICAS DE R²

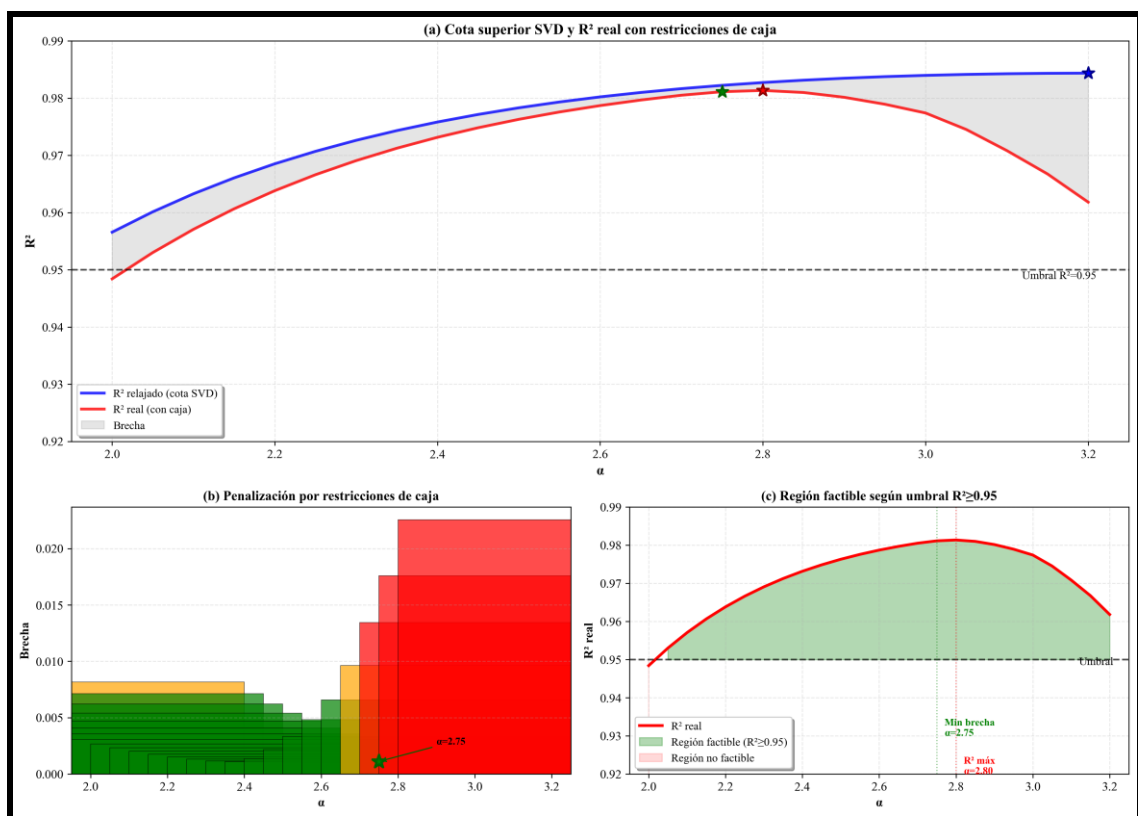
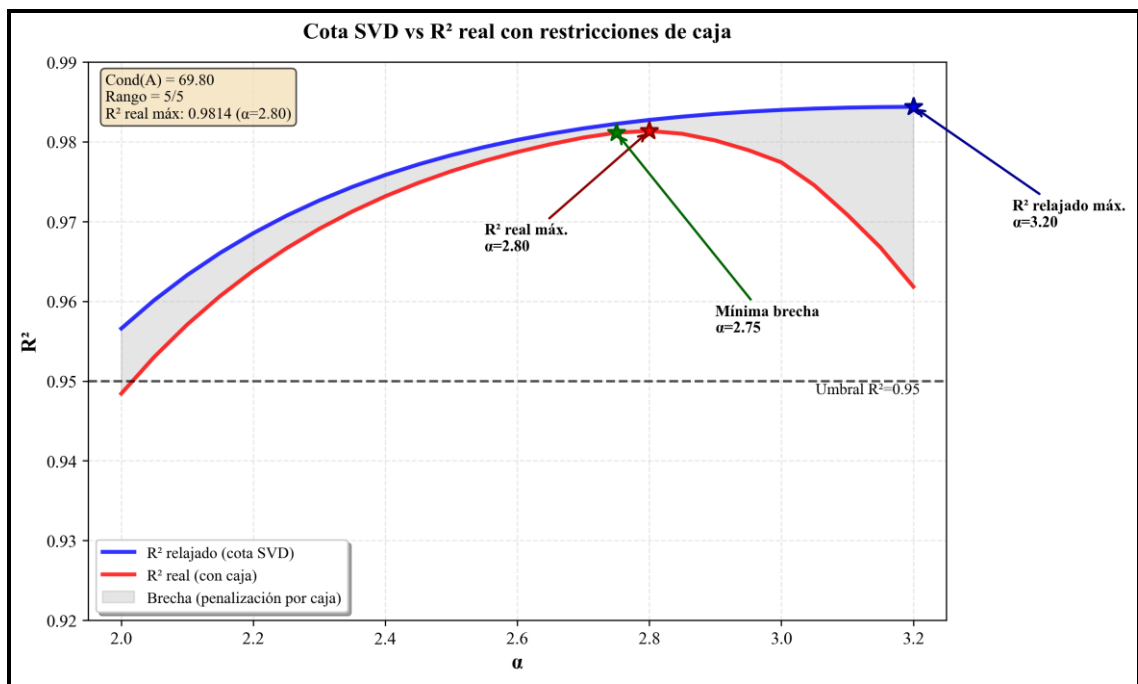
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Método	Mínimo	Máximo	Promedio
Lagrange	0.94842235	0.98135998	0.97135396
RN real	0.94820094	0.98136080	0.97126940
RN redondeada	0.94752125	0.98124064	0.97087410

La función aproximada será del siguiente tipo:

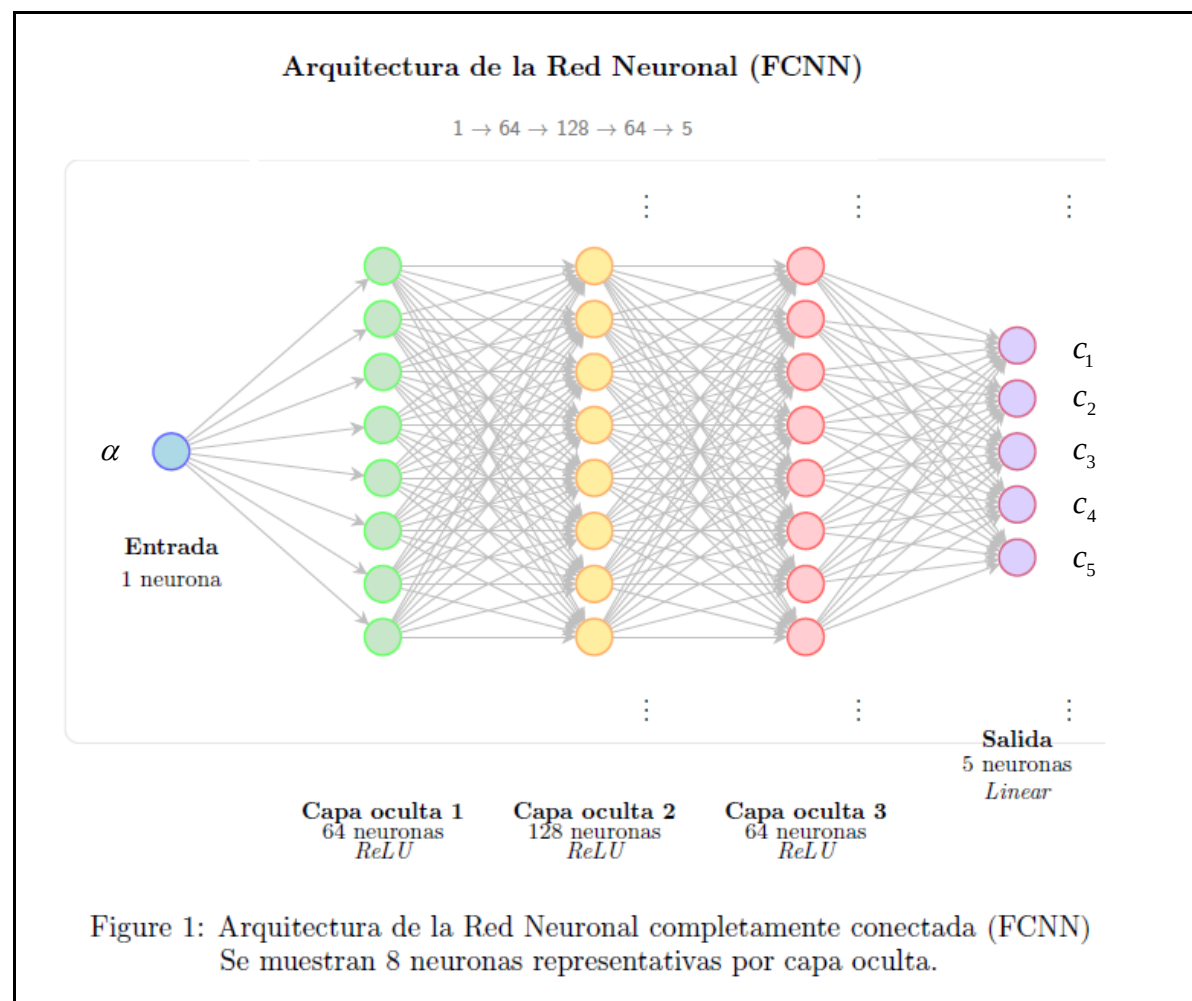
$$y_{aprox}(x) = c_1 \cdot b_1(x) + c_2 \cdot b_2(x) + c_3 \cdot b_3(x) + c_4 \cdot b_4(x) + c_5 \cdot b_5(x)$$

Donde los c_i son los coeficientes dados por los diferentes métodos de aproximación para cada valor de α correspondiente de la la función objetivo, y donde los $b_i(x)$ son las distintas curvas base.



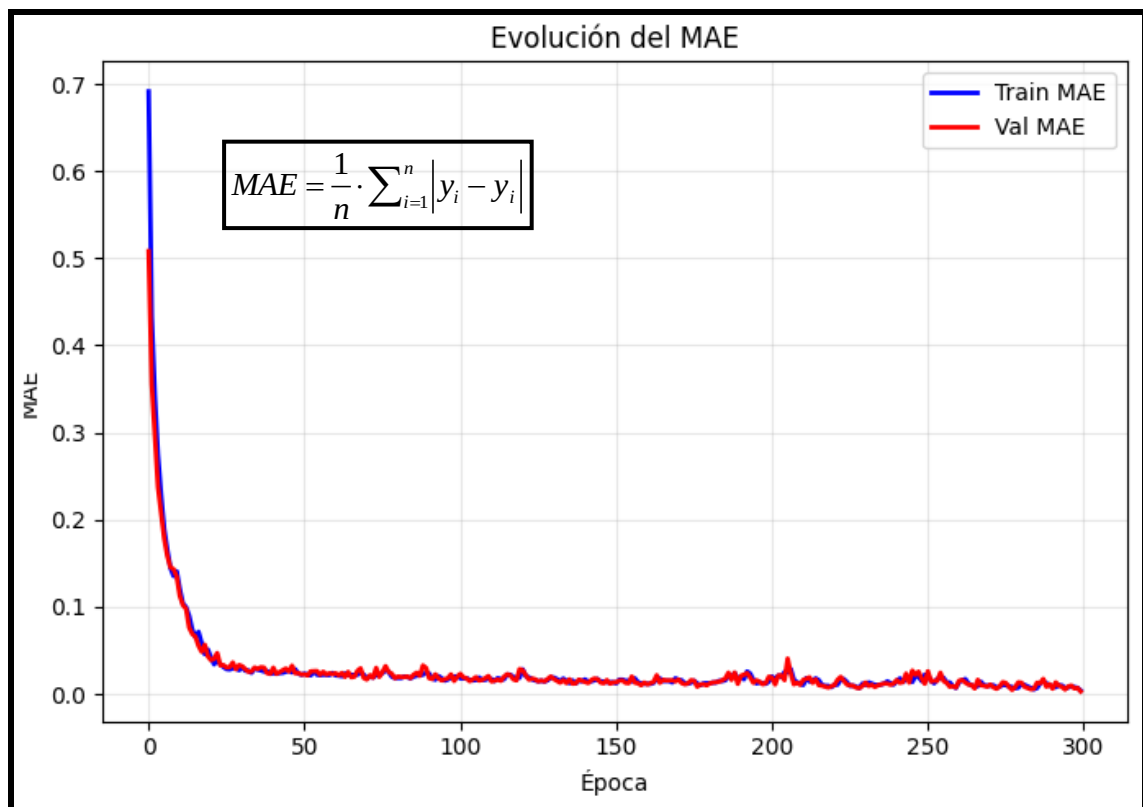
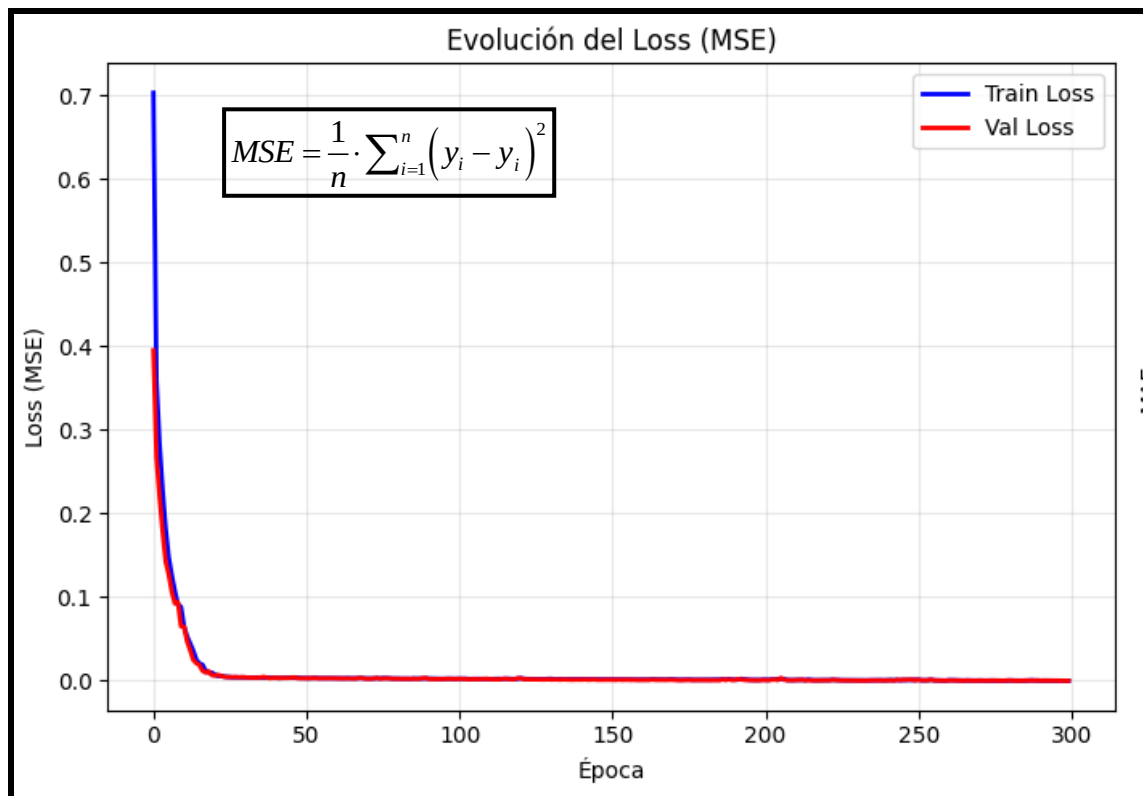
Arquitectura de la Red Neuronal

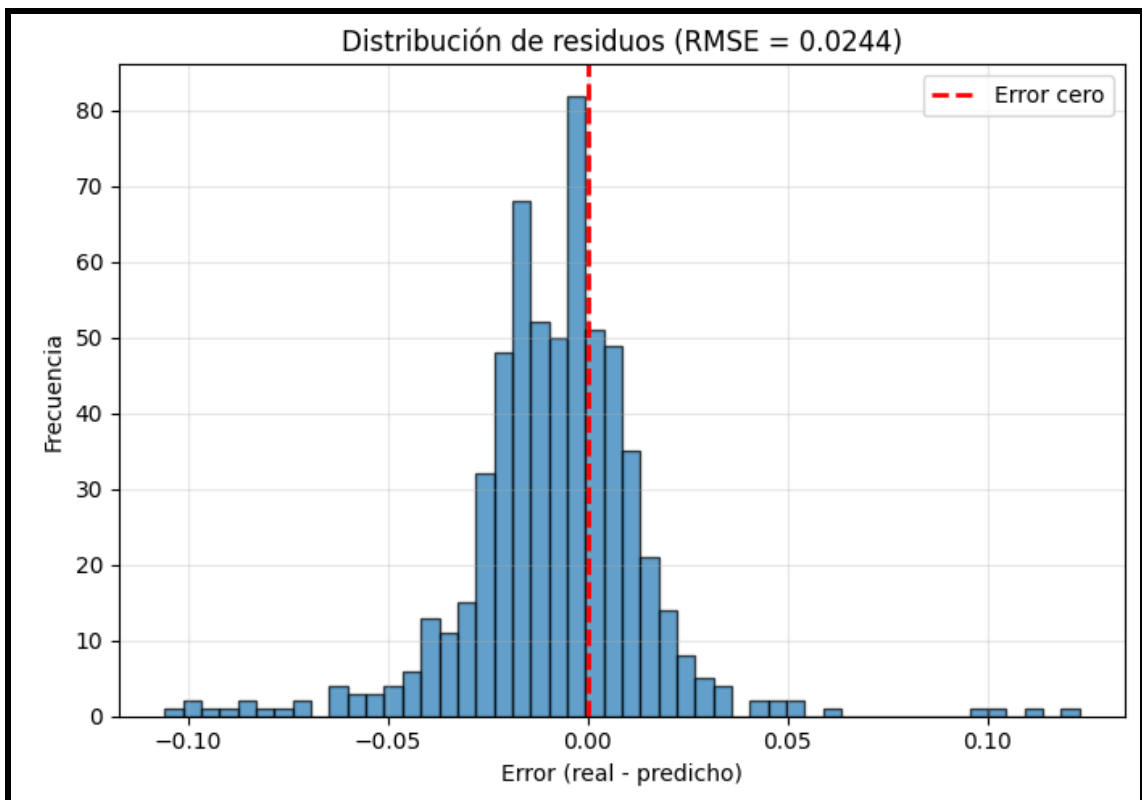
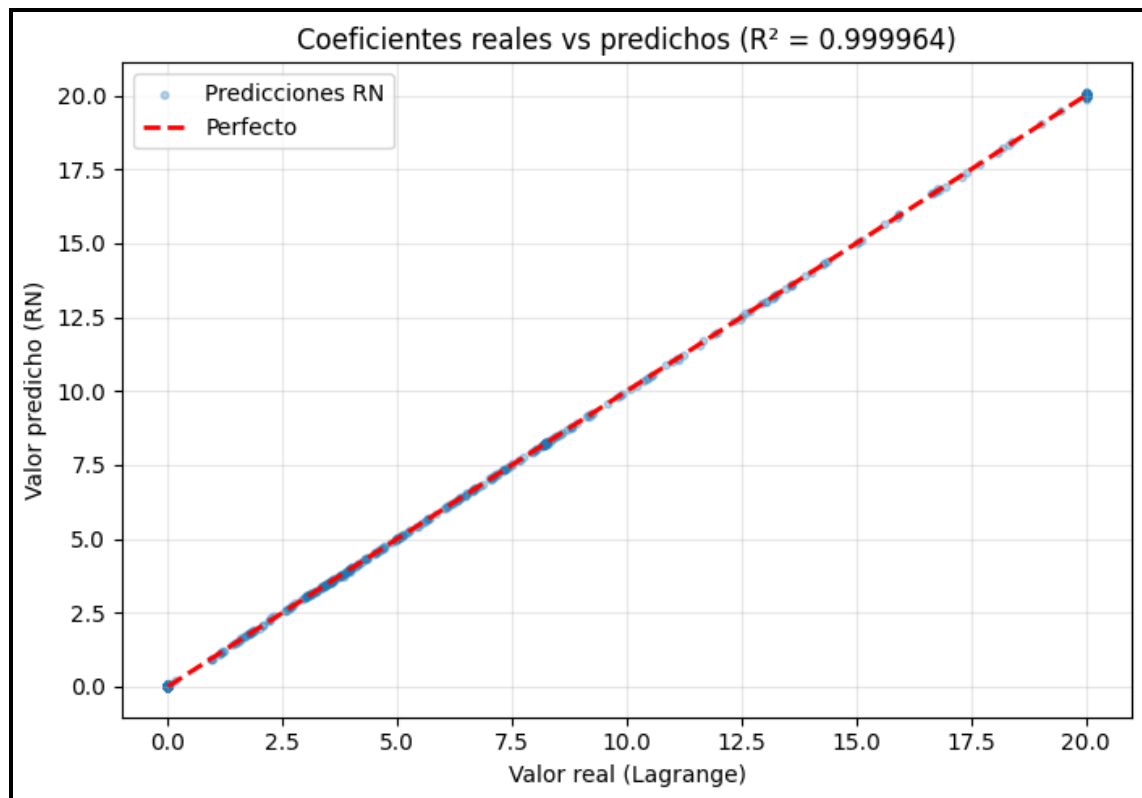
```
model = Sequential([
    Dense(64, activation='relu', input_shape=(1,)), # Capa 1: Entrada → 64 neuronas
    Dense(128, activation='relu'), # Capa 2: 64 → 128 neuronas
    Dense(64, activation='relu'), # Capa 3: 128 → 64 neuronas
    Dense(5, activation='linear') # Capa 4 (salida): 64 → 5 neuronas
])
```

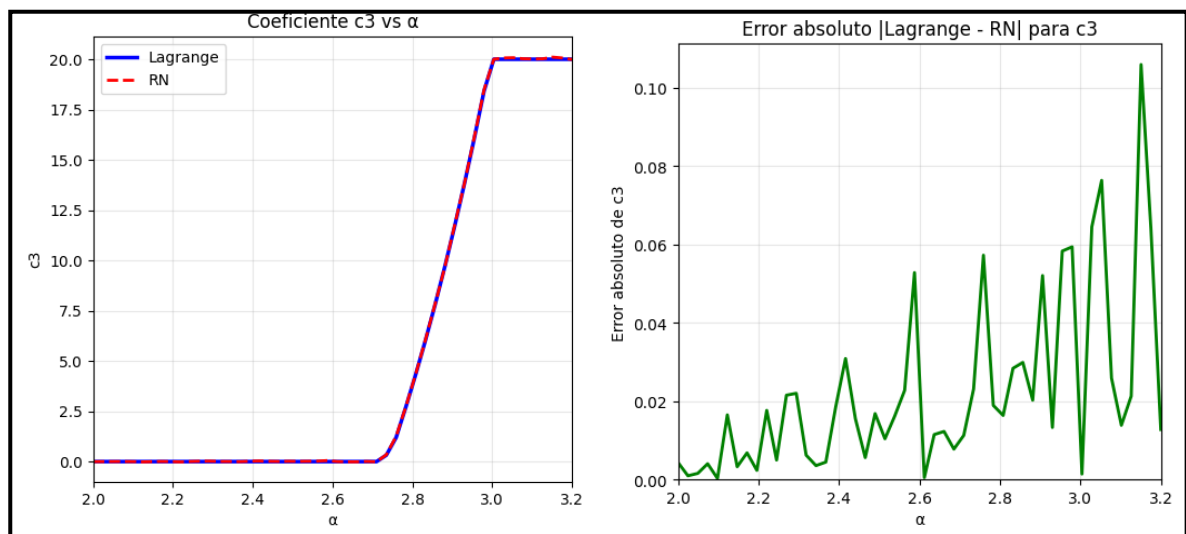
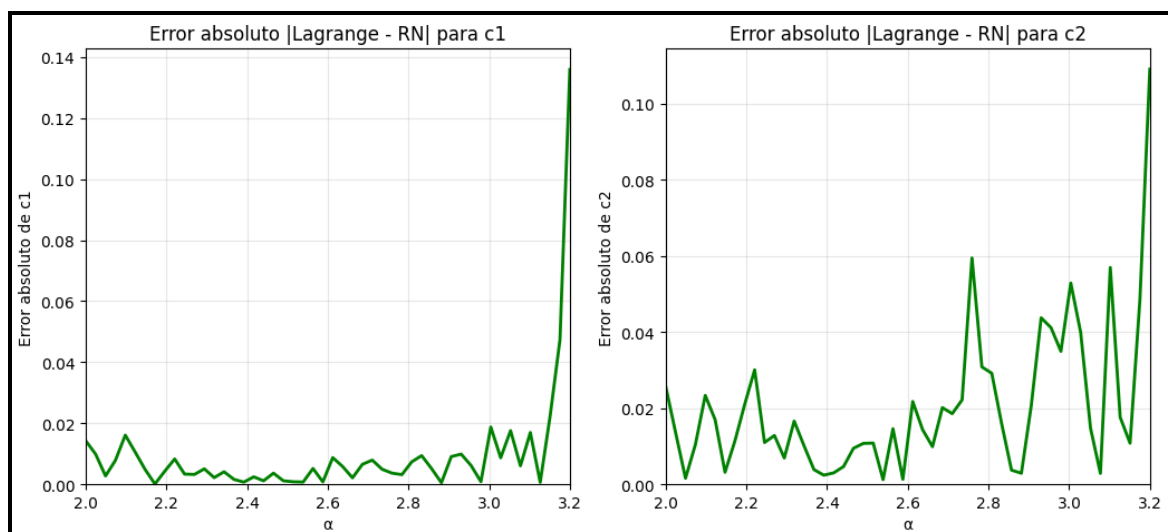
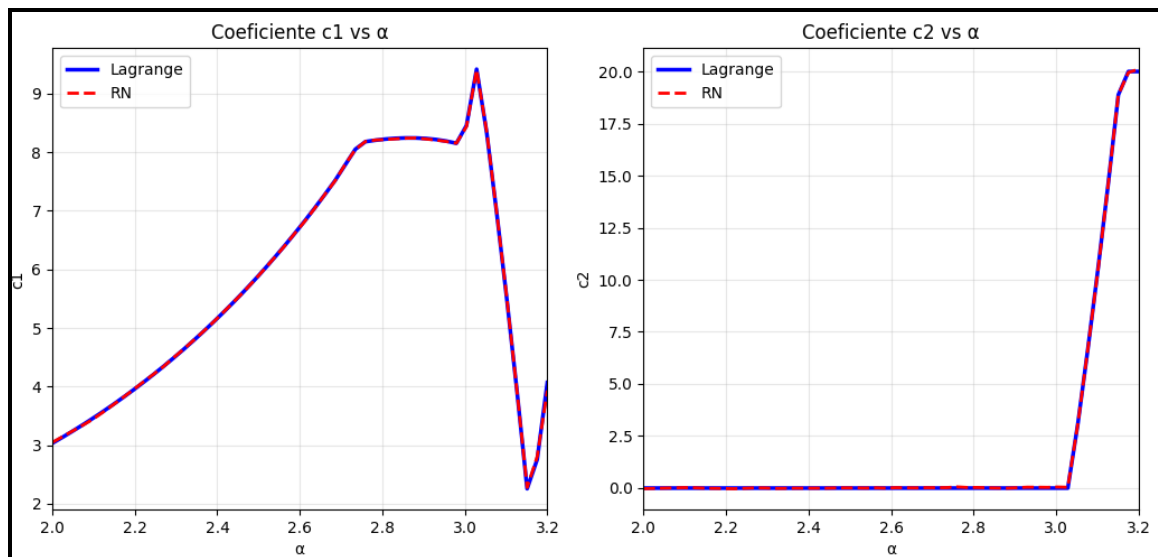


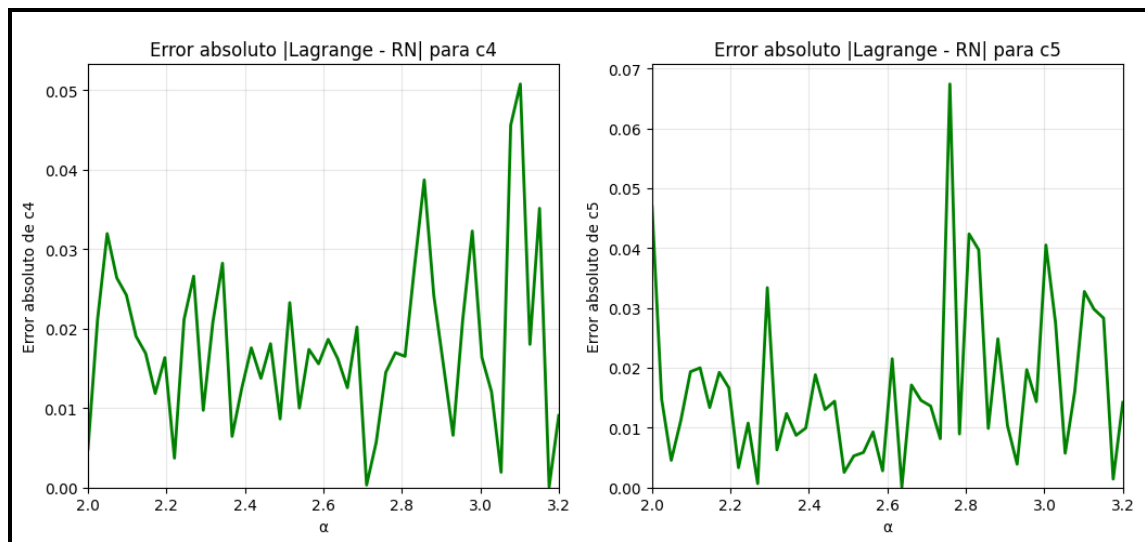
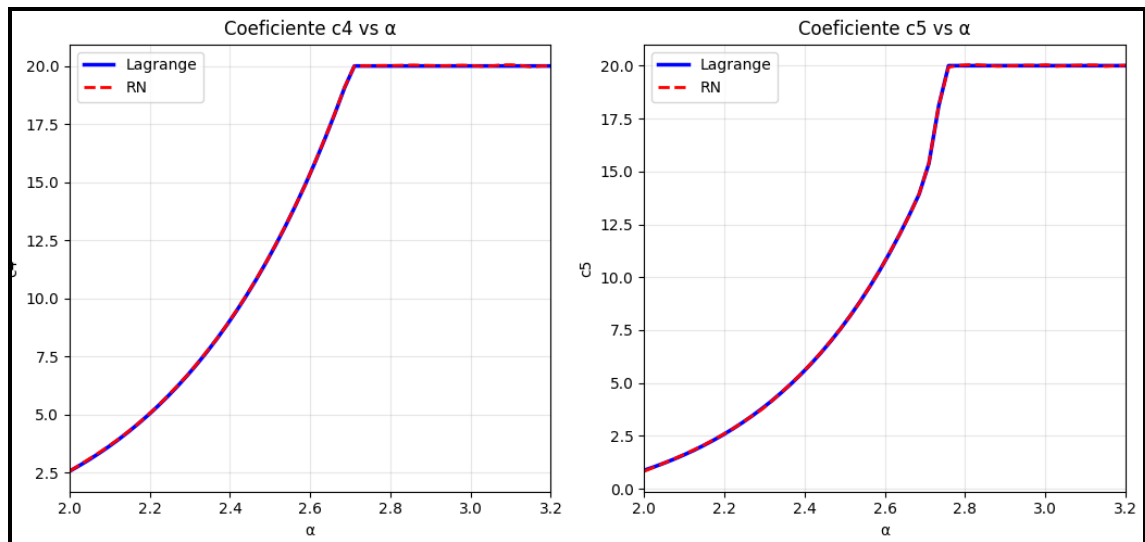
Métricas del entrenamiento y validación de la Red Neuronal

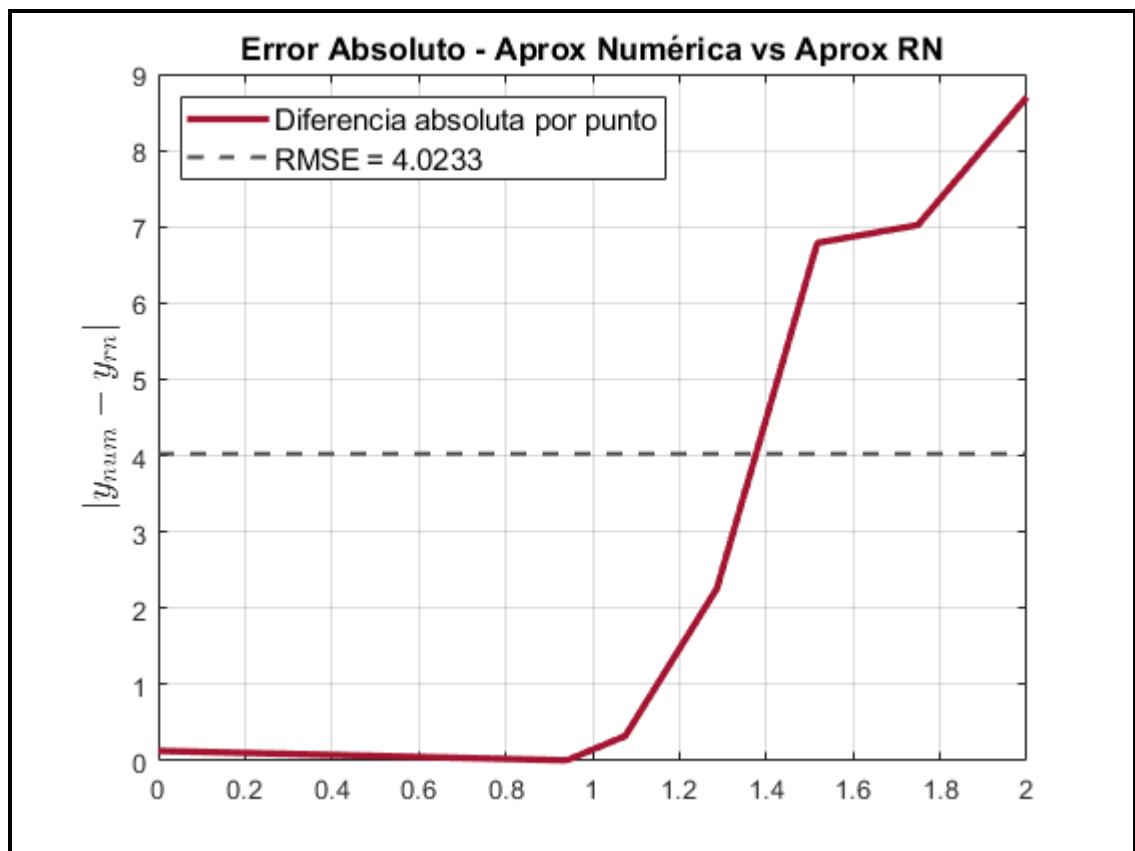
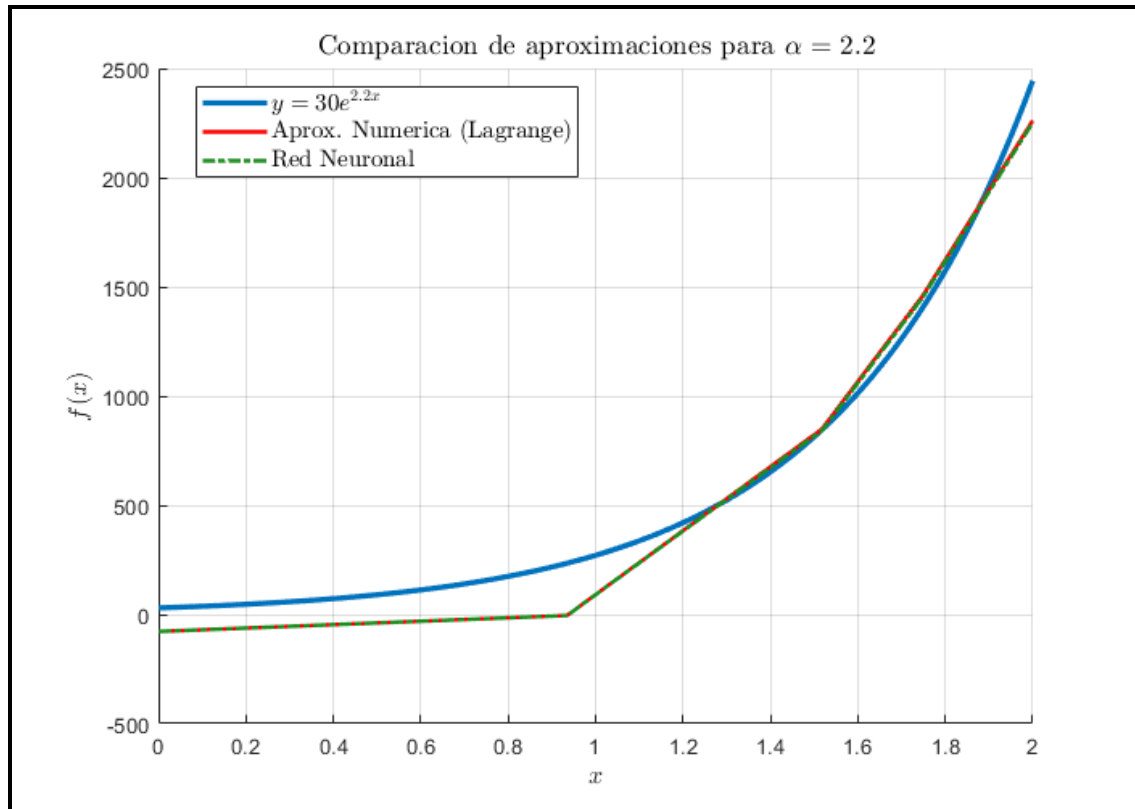
No hay overfitting ya que las pérdidas en el entrenamiento y en el test disminuyen

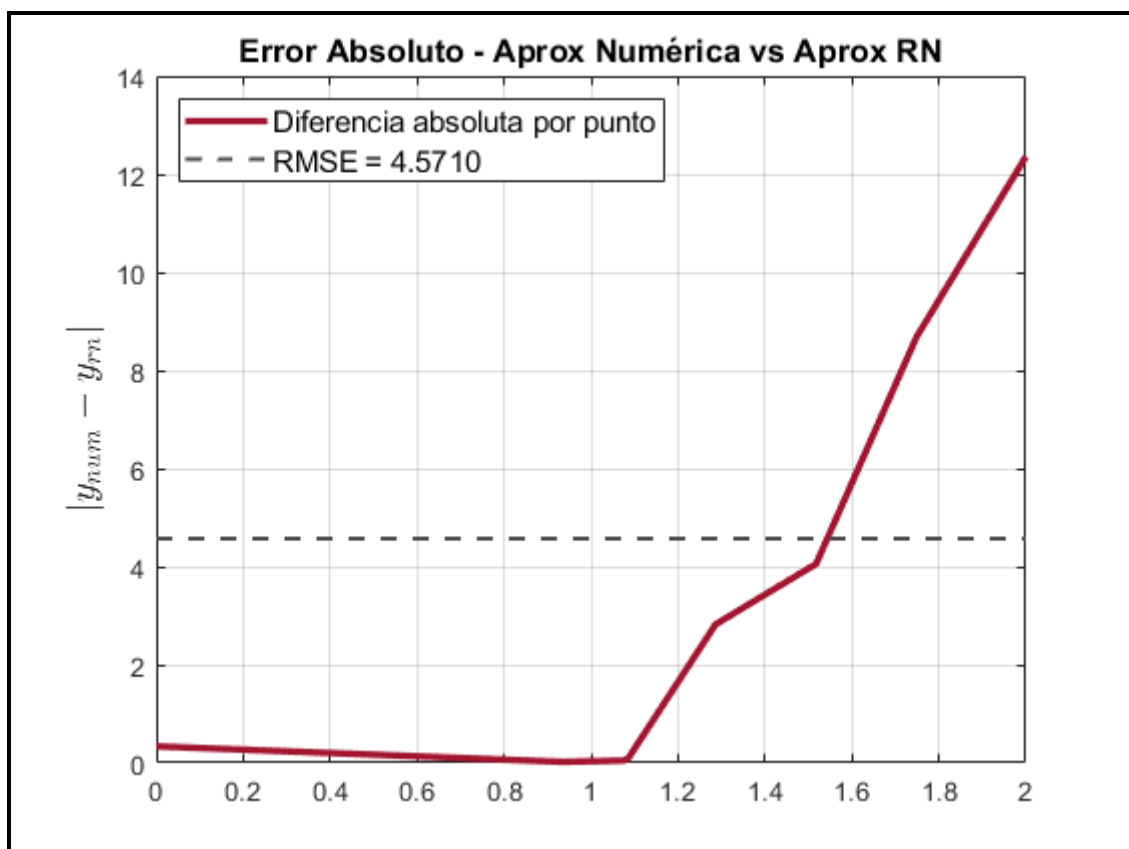
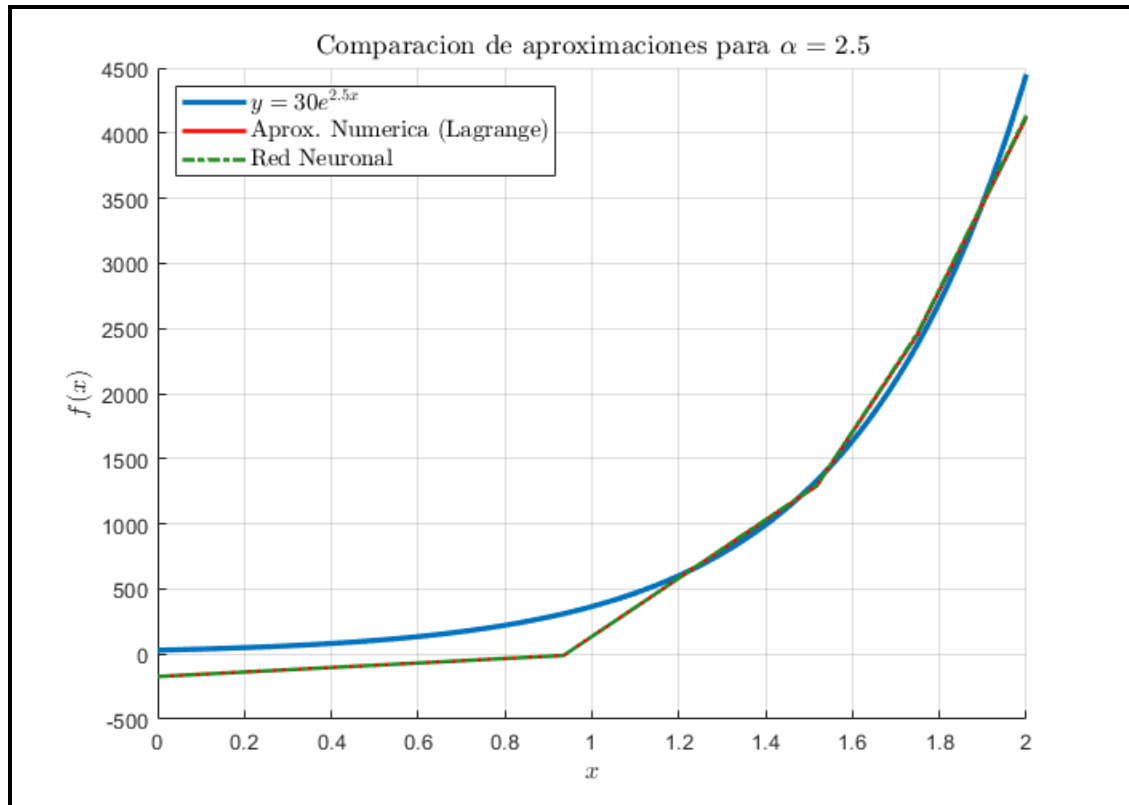


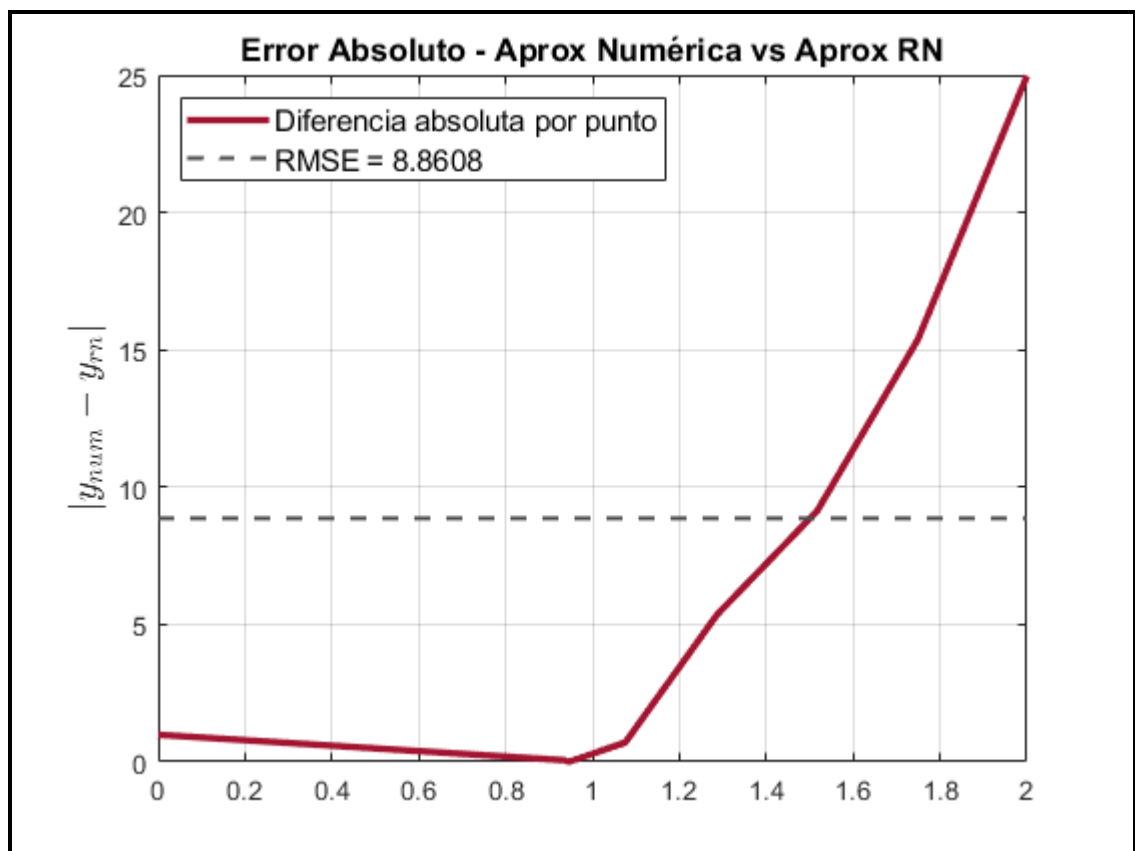
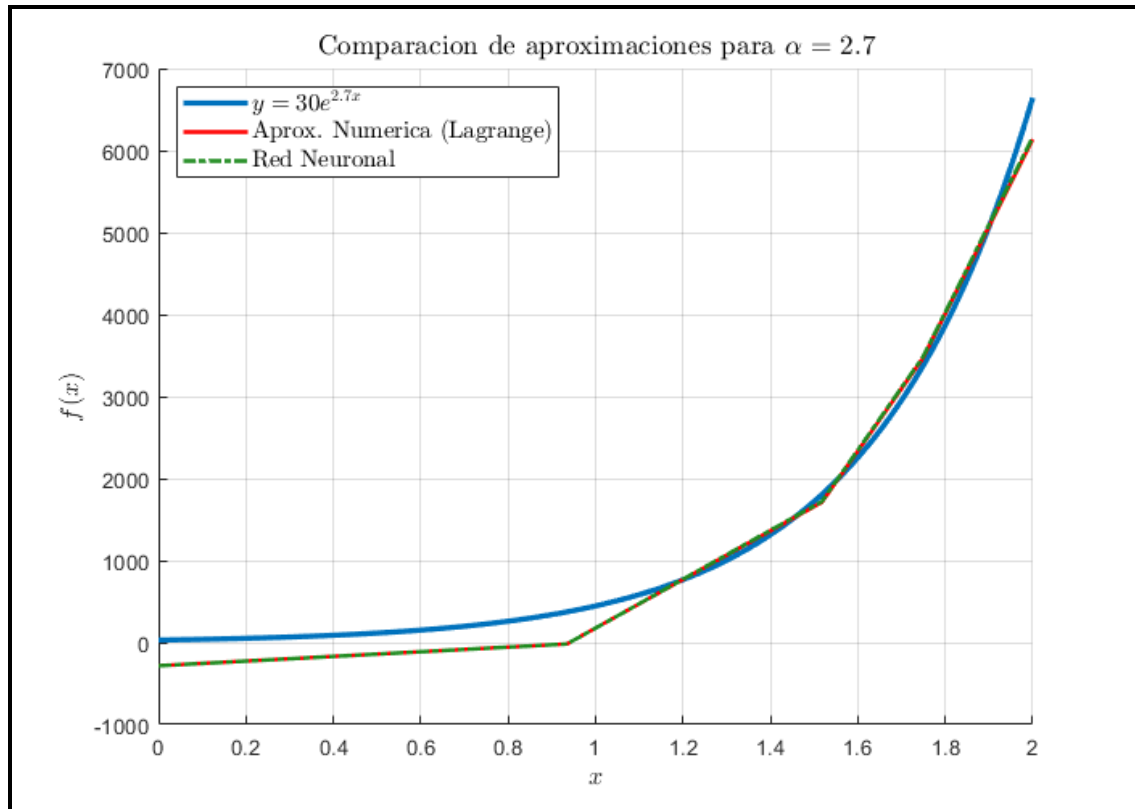


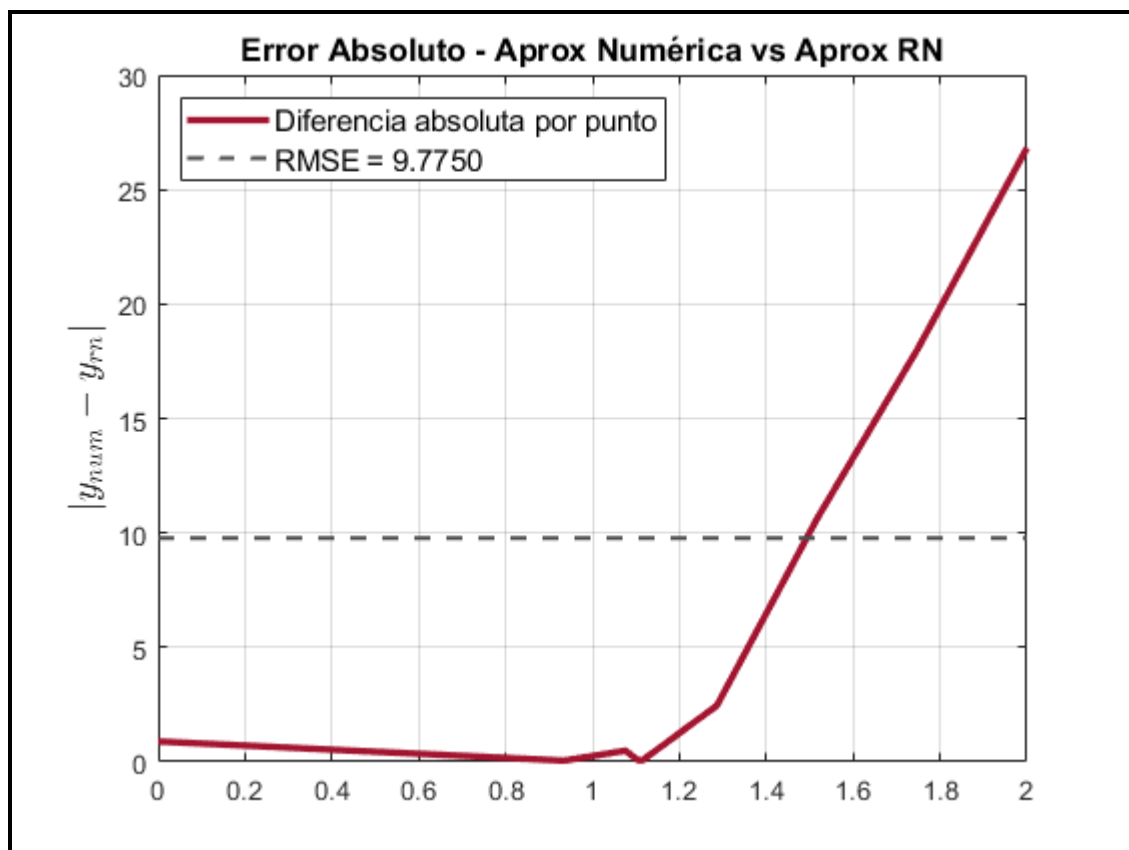
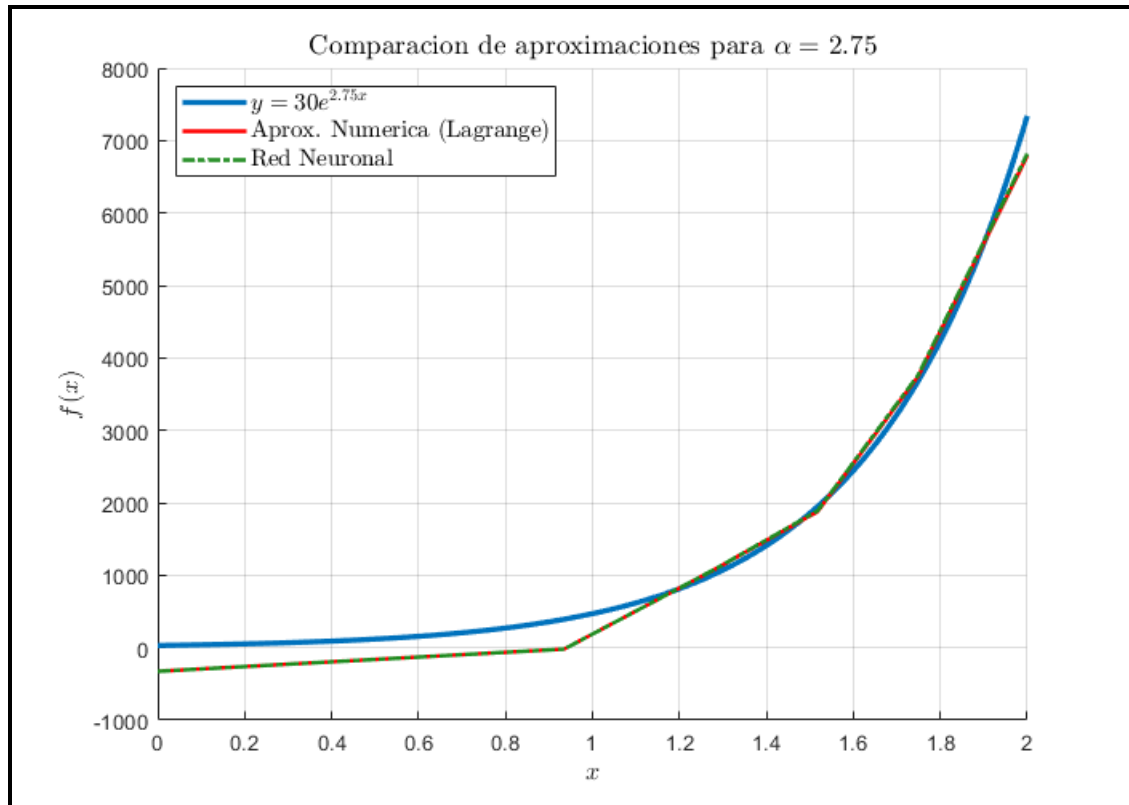


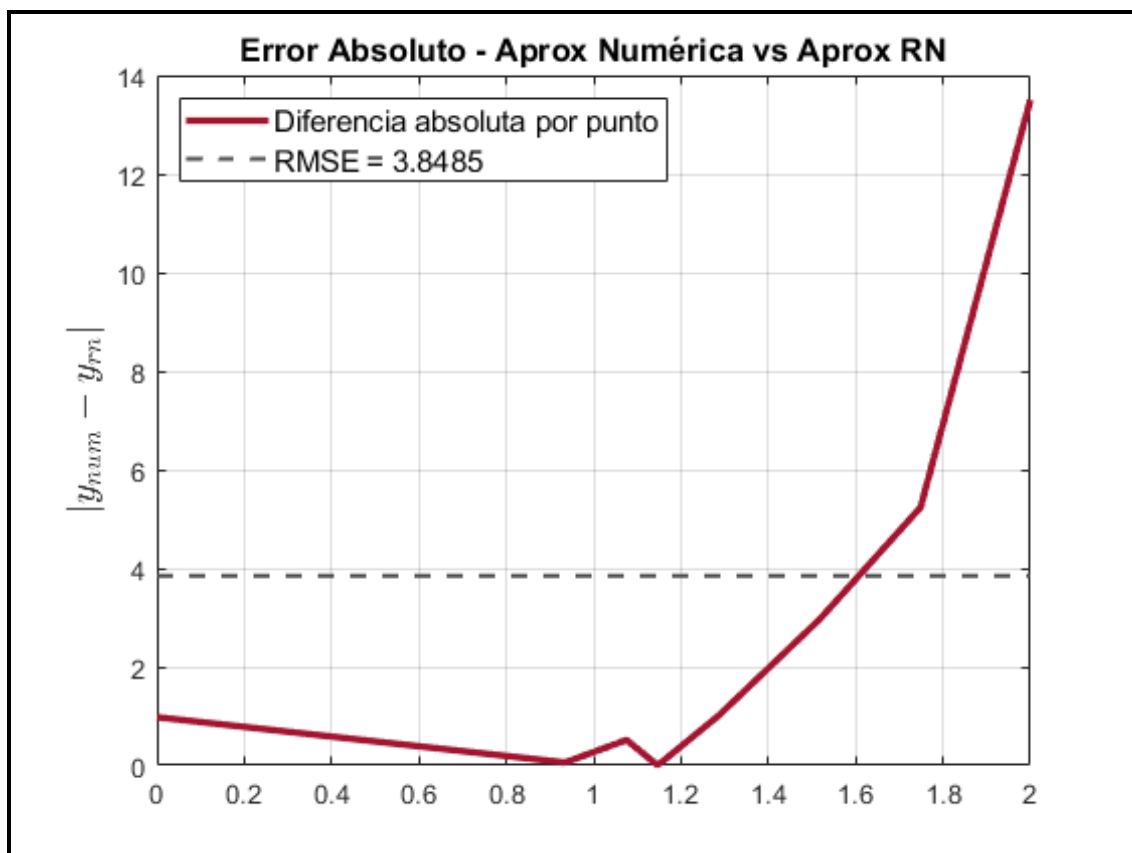
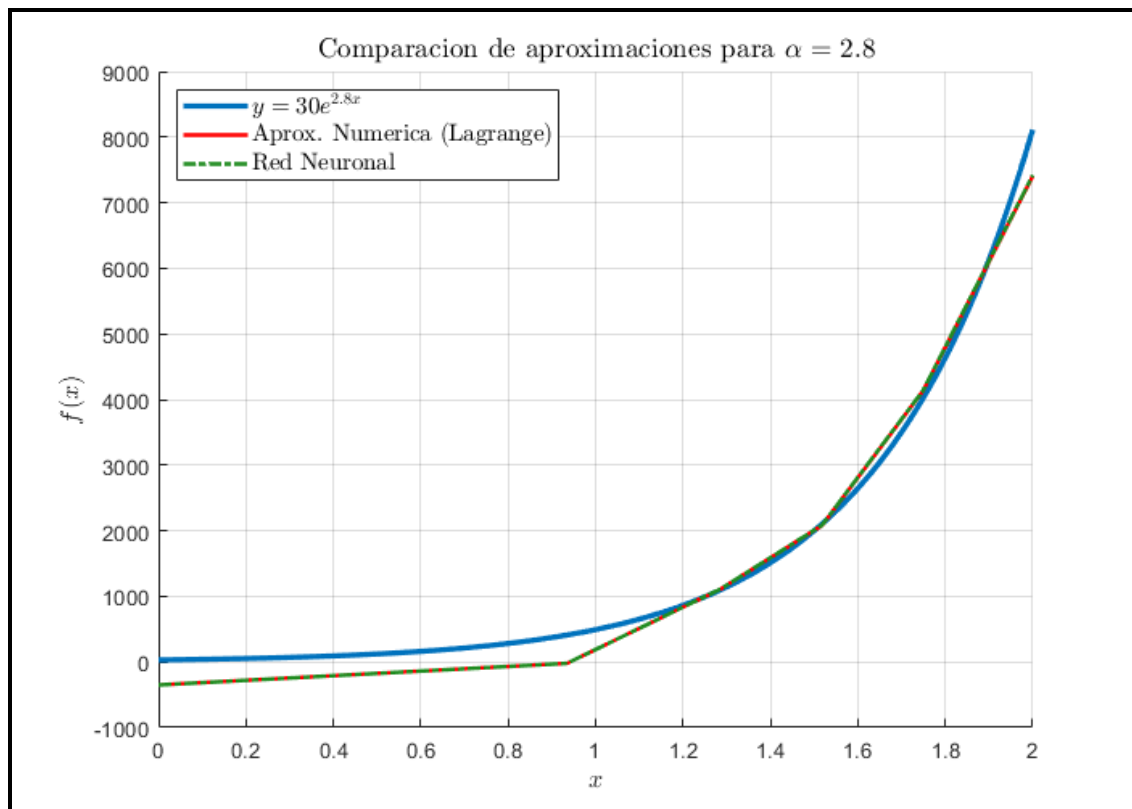


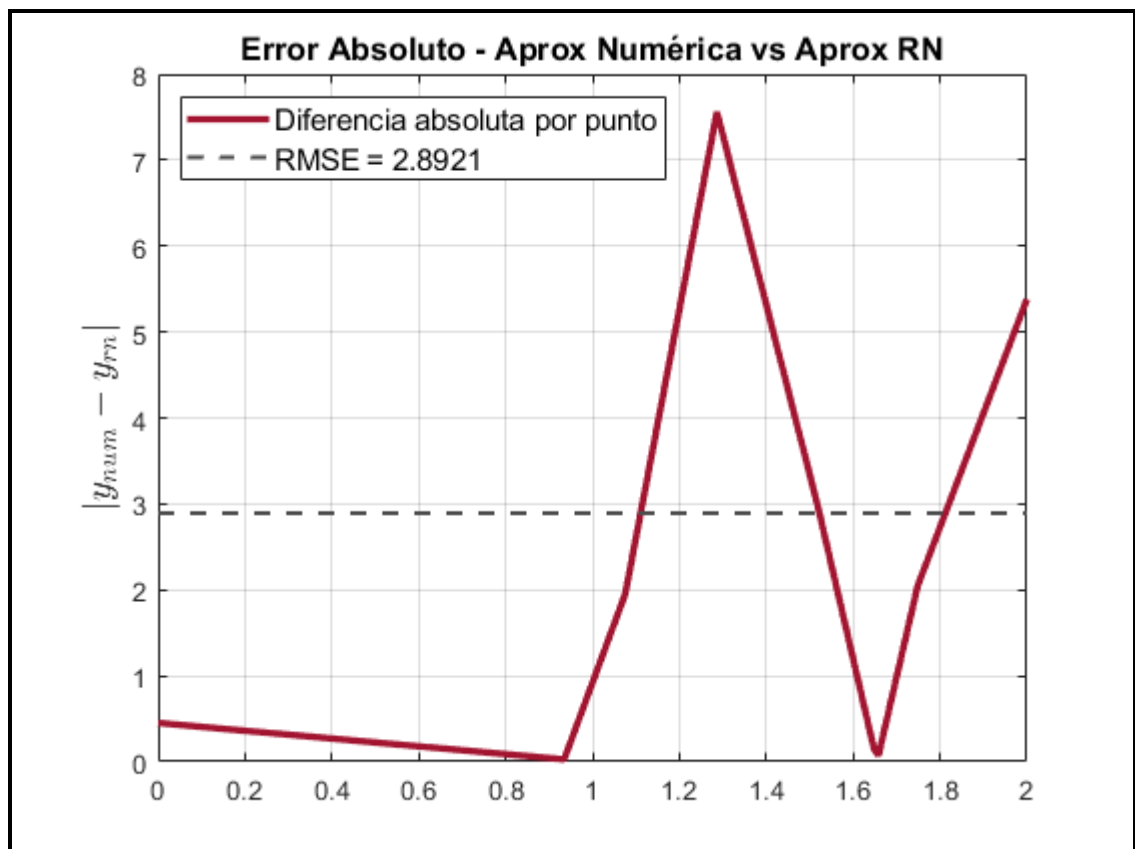
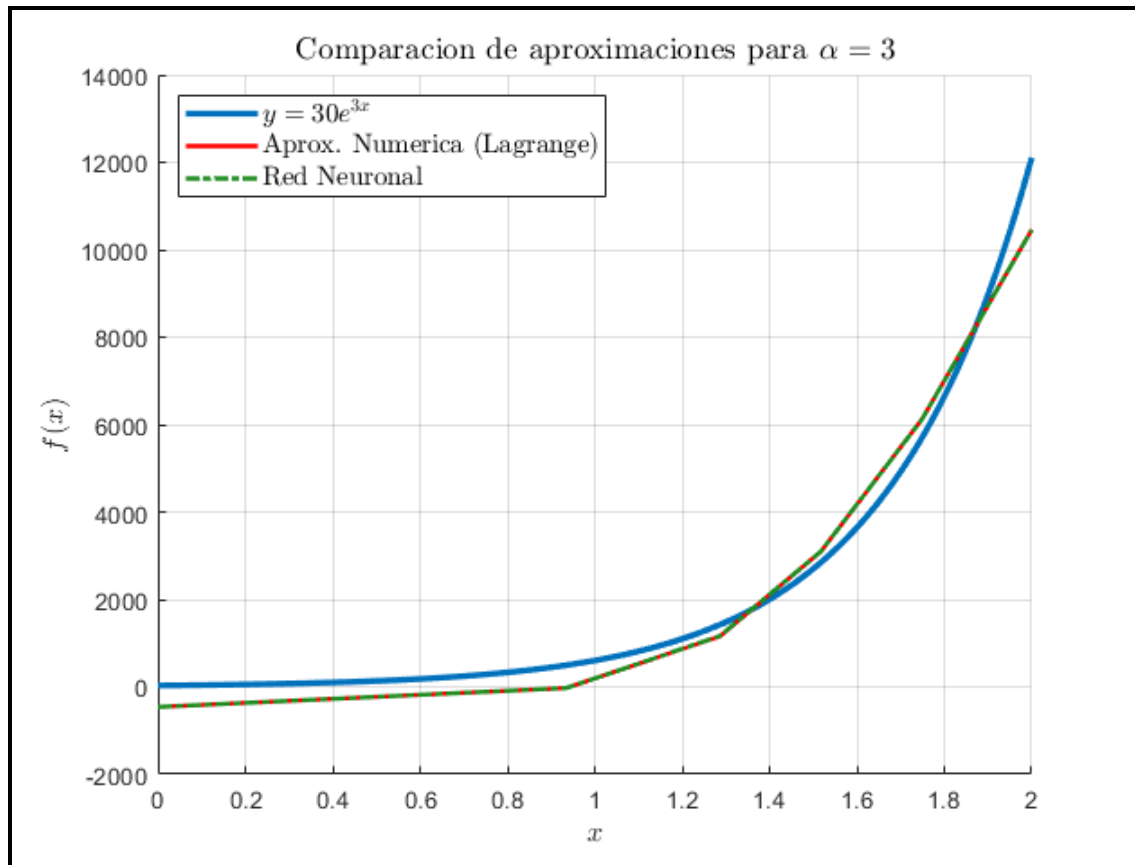


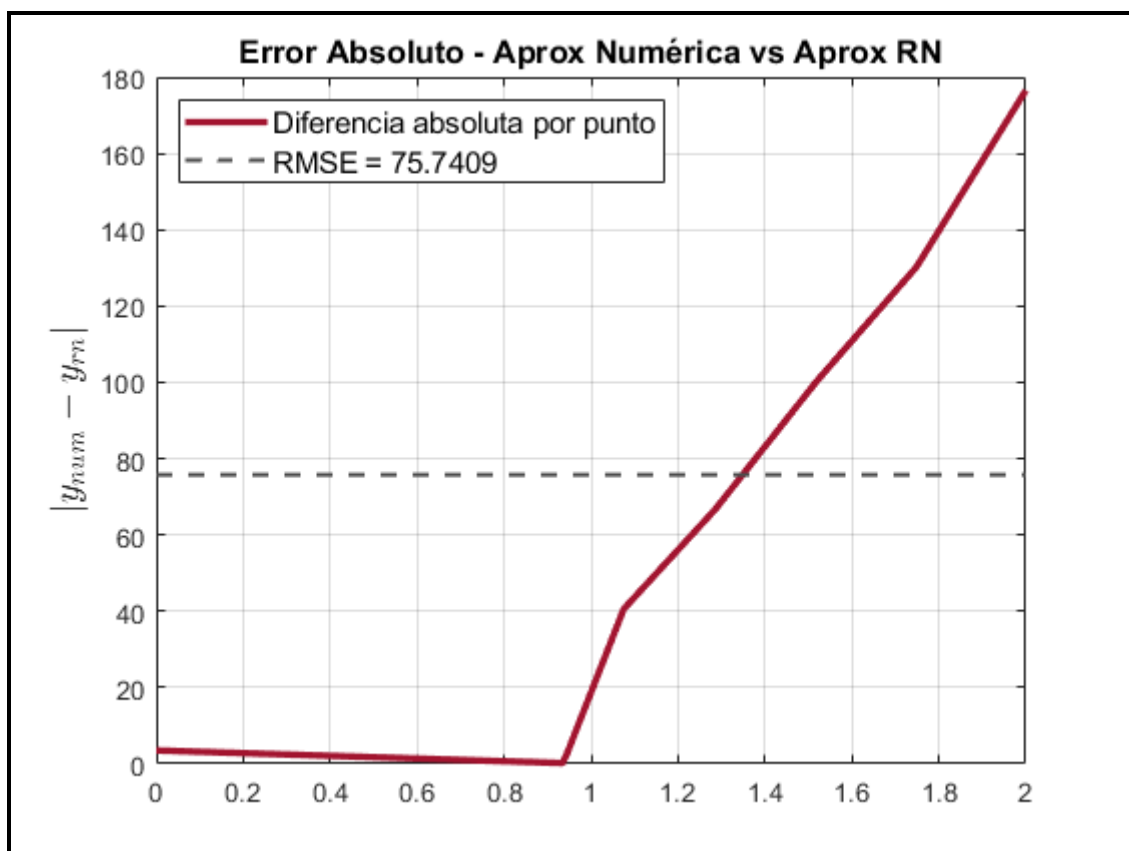
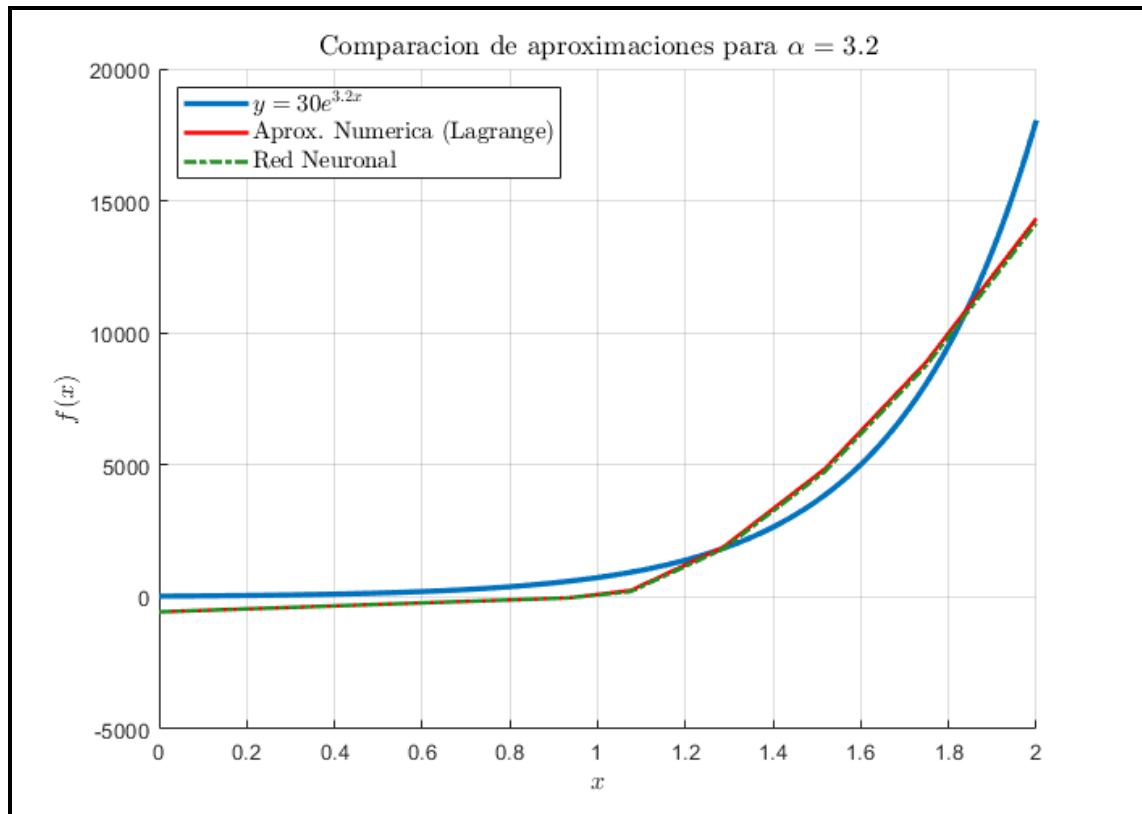


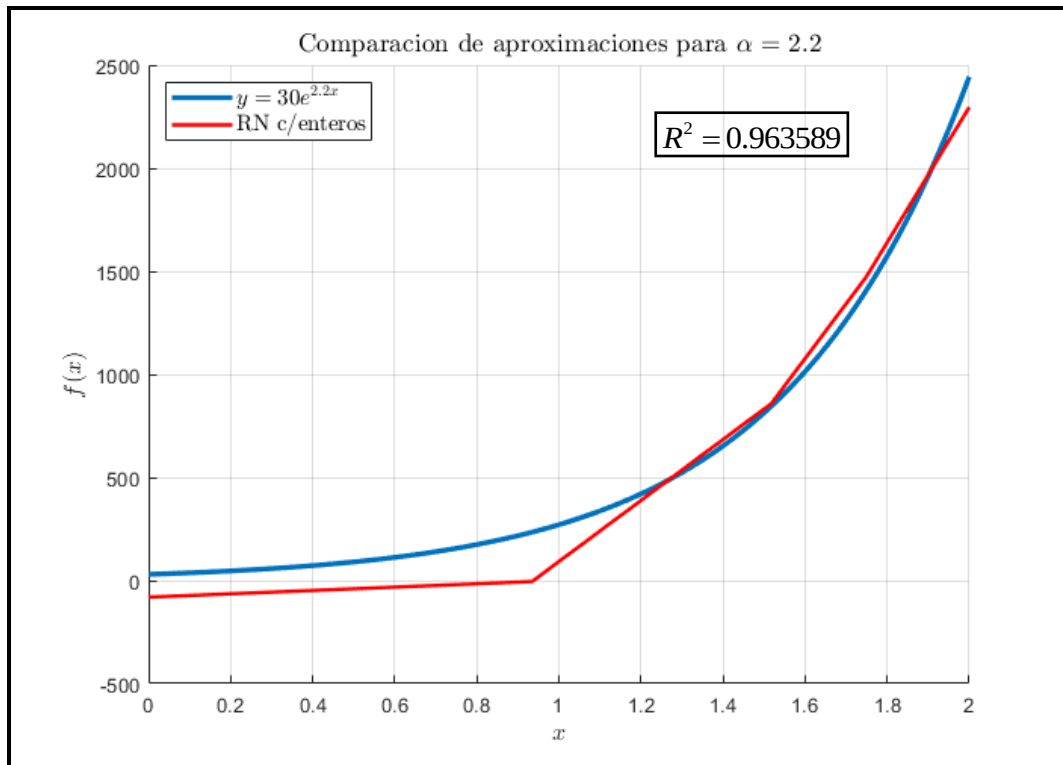




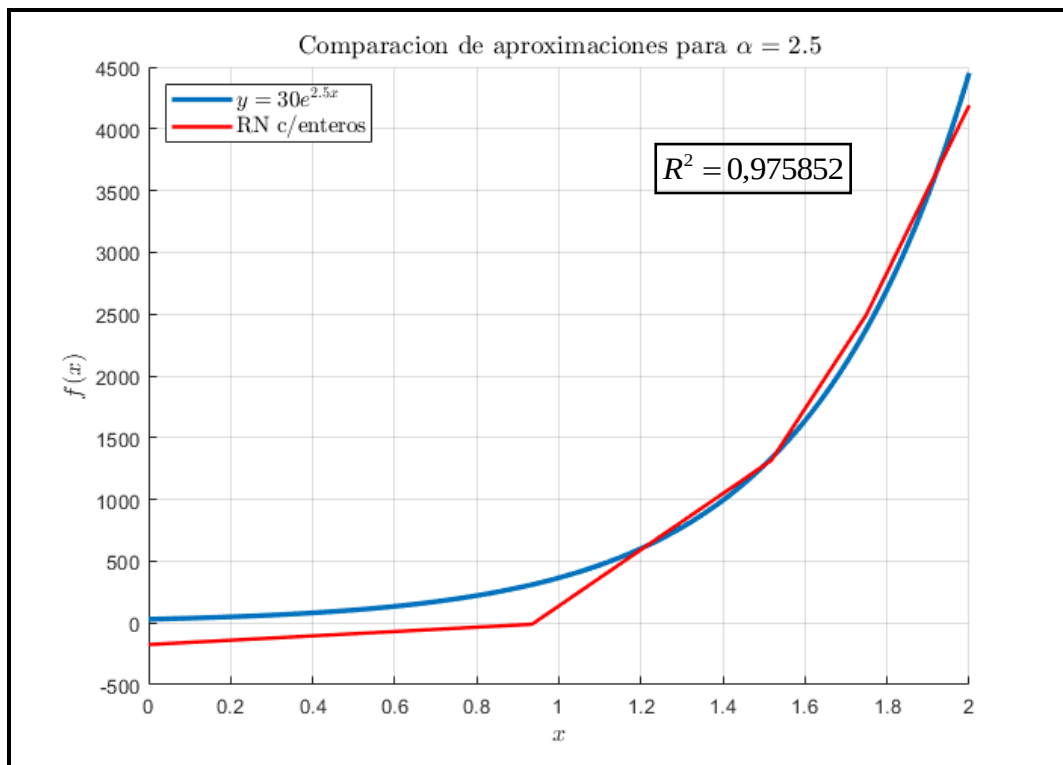




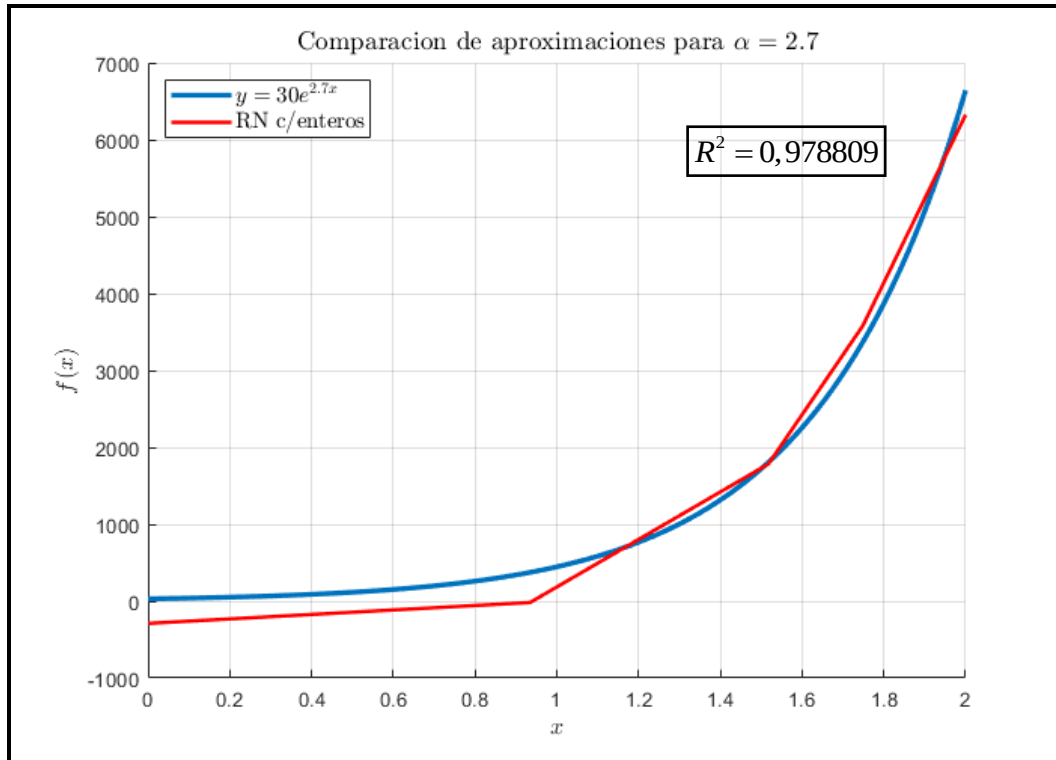




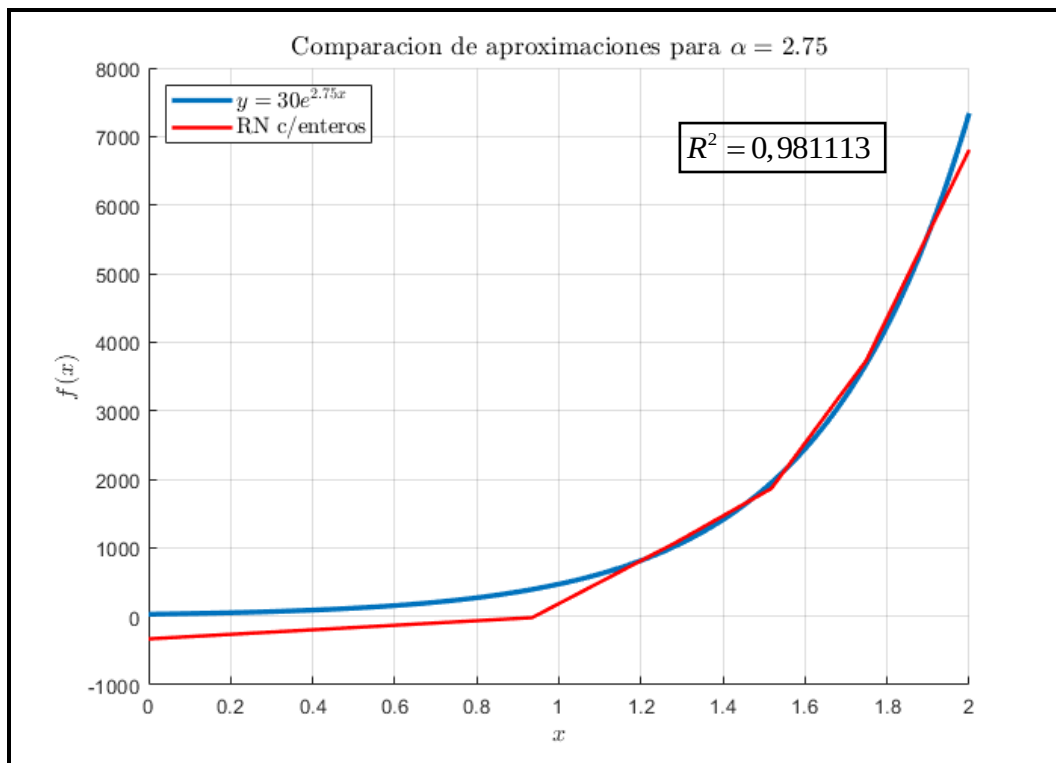
$$y_{aprox} = 4 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 5 \cdot b_4(x) + 3 \cdot b_5(x)$$



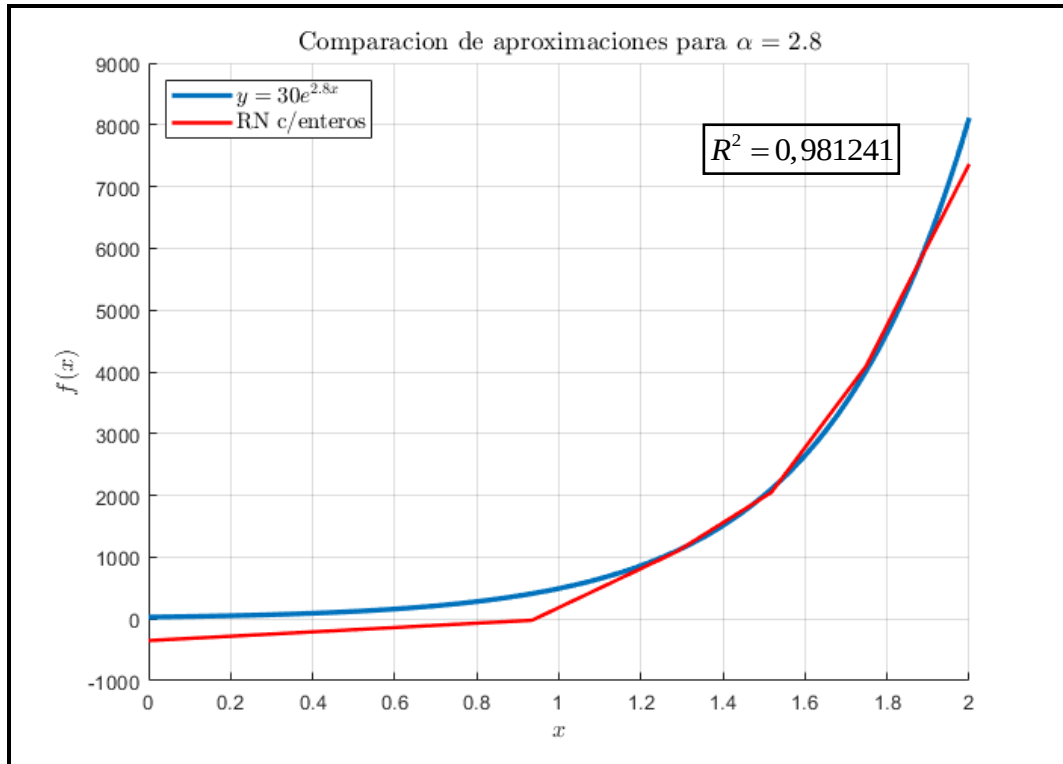
$$y_{aprox} = 6 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 12 \cdot b_4(x) + 8 \cdot b_5(x)$$



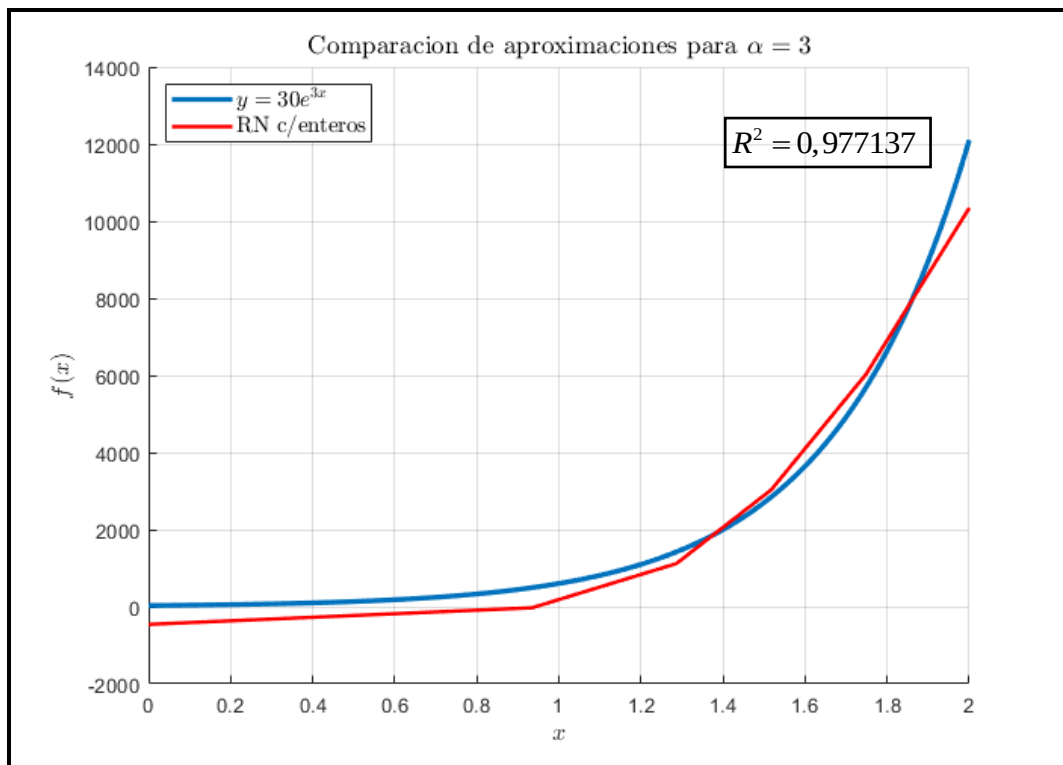
$$y_{aprox} = 8 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 20 \cdot b_4(x) + 15 \cdot b_5(x)$$



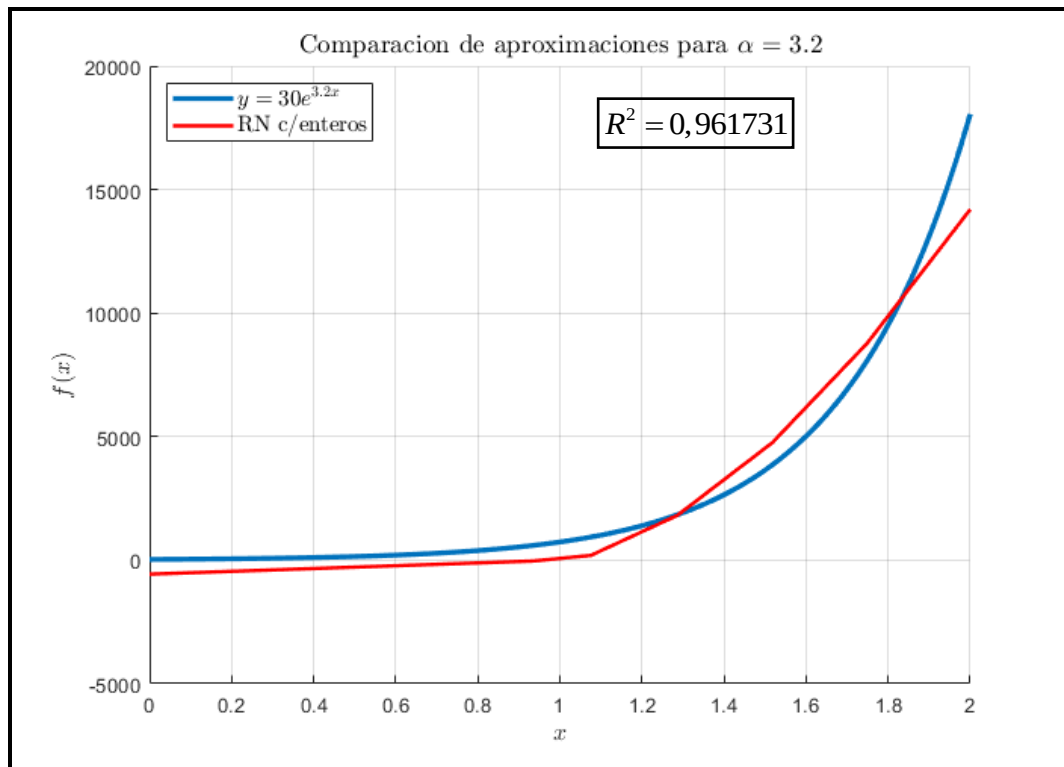
$$y_{aprox} = 8 \cdot b_1(x) + 0 \cdot b_2(x) + 1 \cdot b_3(x) + 20 \cdot b_4(x) + 20 \cdot b_5(x)$$



$$y_{aprox} = 8 \cdot b_1(x) + 0 \cdot b_2(x) + 4 \cdot b_3(x) + 20 \cdot b_4(x) + 20 \cdot b_5(x)$$



$$y_{aprox} = 8 \cdot b_1(x) + 0 \cdot b_2(x) + 20 \cdot b_3(x) + 20 \cdot b_4(x) + 20 \cdot b_5(x)$$



$$y_{aprox} = 3 \cdot b_1(x) + 21 \cdot b_2(x) + 20 \cdot b_3(x) + 20 \cdot b_4(x) + 20 \cdot b_5(x)$$